

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION
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COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

19. Edge Detection (Vertical Sobel)

Apply vertical Sobel:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Matrix:

$$\begin{bmatrix} 10 & 20 & 30 \\ 10 & 20 & 30 \\ 10 & 20 & 30 \end{bmatrix}$$

- (a) Interpret edge detection objective.
- (b) Justify operator choice.
- (c) Apply step-by-step.
- (d) Compute center value.
- (e) Interpret edge direction.

ANS:

Given Matrix:

$$\begin{bmatrix} 10 & 20 & 30 \\ 10 & 20 & 30 \\ 10 & 20 & 30 \end{bmatrix}$$

Vertical Sobel Kernel:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

(a) Interpret edge detection objective

Edge detection identifies locations where image intensity changes sharply. It helps detect object boundaries and shapes.

(b) Justify operator choice

The **Vertical Sobel operator** detects **vertical edges** by measuring intensity changes from left to right. It also gives more weight to the center row, making it less sensitive to noise.

(c) Apply step-by-step

Multiply corresponding elements.

Image	Kernel	Product
10	-1	-10
20	0	0
30	1	30
10	-2	-20
20	0	0
30	2	60
10	-1	-10
20	0	0
30	1	30

Now add all products:

$$\begin{aligned} & -10 + 0 + 30 - 20 + 0 + 60 - 10 + 0 + 30 \\ & = 80 \end{aligned}$$

(d) Compute center value

80

(e) Interpret edge direction

Since the response is **positive and large**, there is a **strong vertical edge**, with brightness increasing from **left to right**.

20. Prewitt Edge Detection

- (a) Interpret Prewitt operator purpose.
- (b) Justify use over Sobel.
- (c) Apply kernel:
[−1 0 1, −1 0 1, −1 0 1]
- (d) Compute output.
- (e) Interpret result.

ANS:

Kernel:

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 20 & 30 \\ 10 & 20 & 30 \\ 10 & 20 & 30 \end{bmatrix}$$

(a) Interpret Prewitt operator purpose

The Prewitt operator detects image edges by estimating intensity gradients. It is mainly used for detecting vertical and horizontal edges.

(b) Justify use over Sobel

- Simpler and faster computation.
- Equal weights make implementation easier.
- Suitable when computational cost is more important than noise reduction.

(c) Apply kernel

Multiply corresponding values.

Image Kernel Product

10	-1	-10
20	0	0
30	1	30
10	-1	-10
20	0	0
30	1	30
10	-1	-10
20	0	0
30	1	30

Sum:

$$\begin{aligned} & -10 + 0 + 30 - 10 + 0 + 30 - 10 + 0 + 30 \\ & = 60 \end{aligned}$$

(d) Compute output

60

(e) Interpret result

The positive value **60** indicates a **vertical edge**, but the response is weaker than Sobel because Prewitt does not emphasize the center row.

21. Region Splitting

Pixel set: [40, 45, 100, 105]

- (a) Interpret segmentation problem.
- (b) Justify splitting method.
- (c) Apply threshold 80.
- (d) Compute regions.
- (e) Interpret separation.

ANS:

Pixel set:

[40, 45, 100, 105]

Threshold = **80**

(a) Interpret segmentation problem

Segmentation divides an image into meaningful regions by grouping similar pixel intensities.

(b) Justify splitting method

Region splitting separates pixels based on a threshold. Pixels with significantly different intensity values are placed into different regions.

(c) Apply threshold 80

- Pixels $< 80 \rightarrow$ Region 1
- Pixels $\geq 80 \rightarrow$ Region 2

(d) Compute regions

Region 1

[40, 45]

Region 2

[100, 105]

(e) Interpret separation

The threshold clearly separates the pixels into **two distinct regions**:

- **Region 1:** Low-intensity (darker) pixels
- **Region 2:** High-intensity (brighter) pixels

This segmentation effectively distinguishes two different image areas based on intensity.

RUBRICS FOR CALCULATION:

Problem Understanding (20)	Method Selection (20)	Solution Process (25)	Accuracy of Calculation (20)	Interpretation of Results (15)